

I.2 — Albanese Varieties and Albanese Mappings

Fibrations in Birational Geometry Reading Notes • Summer 2026

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Big picture

The aim of this note is to give a self-contained introduction to Albanese varieties and Albanese mappings, with a view towards minimal model theory and the classification of certain varieties. We first give an elementary introduction to Albanese varieties and the Albanese mapping. We then turn to two research topics: the existence of good minimal models for varieties with maximal Albanese dimension ([Section 5](#)), and Demailly–Peternell–Schneider’s classification of compact Kähler manifolds with nef tangent bundle ([Section 4](#)). We end the note by pointing out some further directions.

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1 Albanese varieties

1.1 Construction over a compact Kähler manifold

Let us first construct the Albanese torus. The following observation is obvious: on a compact Kähler manifold holomorphic forms are automatically closed.

Lemma 1.1. *Let X be a compact Kähler manifold with Kähler metric ω . If $\alpha \in H^0(X, \Omega_X^p)$ is a holomorphic p -form, then α is closed.*

Proof. By definition $\bar{\partial}\alpha = 0$. By type reasons the formal adjoint satisfies $\bar{\partial}^*\alpha = 0$ as well, so $\Delta_{\bar{\partial}}\alpha = 0$. The Kähler identities give $\Delta_{\bar{\partial}} = \Delta_{\partial}$, so $\Delta_{\partial}\alpha = 0$. Then

$$0 = (\Delta_{\partial}\alpha, \alpha) = \|\partial\alpha\|^2 + \|\bar{\partial}^*\alpha\|^2 \implies \partial\alpha = 0.$$

Together with $\bar{\partial}\alpha = 0$ this gives $d\alpha = 0$. ■

The following proposition is important for the construction of Albanese torus for compact Kähler manifold.

Proposition 1.2. *Let X be a compact Kähler manifold with Kähler form ω .*

(1) *There is a well-defined homomorphism*

$$\varphi : H_1(X, \mathbb{Z}) \rightarrow (H^0(X, \Omega_X^1))^{\vee}, \quad [\gamma] \mapsto \left(\alpha \mapsto \int_{\gamma} \alpha \right).$$

(2) *The image of φ is a lattice in $(H^0(X, \Omega_X^1))^{\vee}$; hence the quotient is a complex torus of dimension $\dim_{\mathbb{C}} H^0(X, \Omega_X^1)$.*

Proof of (1). By Lemma 1.1 every $\alpha \in H^0(X, \Omega_X^1)$ is closed. If $[\gamma] = [\gamma']$ in $H_1(X, \mathbb{Z})$, there is a singular 2-chain S with $\gamma - \gamma' = \partial S$, so by Stokes' theorem

$$\int_{\gamma - \gamma'} \alpha = \int_{\partial S} \alpha = \int_S d\alpha = 0.$$

Hence $\varphi([\gamma])$ depends only on the homology class $[\gamma]$. ■

Proof of (2). Our proof follows [Huy05, page 135]. We must show that $H_1(X, \mathbb{Z})$ maps under φ to a lattice in $(H^0(X, \Omega_X^1))^{\vee}$. The natural map

$$H^{2n-1}(X, \mathbb{Z}) \rightarrow H^{2n-1}(X, \mathbb{C})$$

has image a lattice, since by the universal coefficient theorem tensoring with $\mathbb{Q}$ kills torsion. Applying Serre duality (SD) and Poincaré duality (PD) gives a commutative diagram

$$\begin{array}{ccc} H^0(X, \Omega_X^1)^{\vee} & \xleftarrow[\cong]{\text{SD}} & H^n(X, \Omega_X^{n-1}) \\ \uparrow \varphi & & \uparrow p \\ H_1(X, \mathbb{Z}) & \xrightarrow[\cong]{\text{PD}} & H^{2n-1}(X, \mathbb{Z}) \end{array}$$

where p is the composite $H^{2n-1}(X, \mathbb{Z}) \rightarrow H^{2n-1}(X, \mathbb{C}) \rightarrow H^{n-1, n}(X)$, whose image is again a lattice (tensoring with $\mathbb{C}$ kills torsion).

The image of φ agrees with the image of p : by the definition of Poincaré duality,

$\int_X \omega \wedge \text{PD}([\gamma]) = \int_\gamma \omega$, while by the definition of Serre duality

$$\text{SD}(\text{PD}([\gamma]))(\omega) := \int_X \text{PD}([\gamma]) \wedge \omega = \int_\gamma \omega = \varphi([\gamma])(\omega).$$

Hence $\text{im}(\varphi) = \text{im}(p)$ is a lattice, of full rank $2 \dim_{\mathbb{C}} H^0(X, \Omega_X^1)$. ■

Remark 1.3 (Lattices). A lattice in a complex vector space V of dimension g is a discrete subgroup $\Lambda \subset V$ of maximal rank, i.e. a free abelian group of rank $2g$; the quotient V/Λ is then a complex torus of dimension g .

Definition 1.4 (Albanese torus, first version). Let X be a compact Kähler manifold. The complex torus

$$\text{Alb}(X) := (H^0(X, \Omega_X^1))^{\vee} / H_1(X, \mathbb{Z})$$

is called the Albanese variety (or Albanese torus) of X ; by Proposition 1.2 it is well defined. Fixing a base point $x_0 \in X$ produces the Albanese mapping (Definition 2.1 below).

2 Albanese mappings

2.1 Construction of Albanese mapping

Definition 2.1 (Albanese mapping, [Huy05, Definition 3.3.7]). Let X be a compact Kähler manifold, fix $x_0 \in X$, and let $\omega_1, \dots, \omega_q$ be a basis of $H^0(X, \Omega_X^1)$. The Albanese mapping of X is

$$\text{alb}_X : X \rightarrow \text{Alb}(X), \quad x \mapsto \left(\int_{x_0}^x \omega_1, \dots, \int_{x_0}^x \omega_q \right) \bmod H_1(X, \mathbb{Z}).$$

Remark 2.2. Since X is compact, the Albanese mapping is proper for trivial reasons. Unfortunately, the Albanese mapping need not be projective in general.

Proposition 2.3. *The Albanese mapping is well defined, i.e., it does not depend on the choice of path from x_0 to x , and a different choice of base point translates alb_X by a constant.*

Proof. Fix the basis $\omega_1, \dots, \omega_q$ of $H^0(X, \Omega_X^1)$ used to define alb_X . By Lemma 1.1, each ω_i is closed.

We first show that it is independent of the path. Let $x \in X$ and let γ, γ' be two paths from

x_0 to x . Then $\sigma := \gamma \cdot (\gamma')^{-1}$ is a closed loop at x_0 , and for each i ,

$$\int_{\gamma} \omega_i - \int_{\gamma'} \omega_i = \int_{\gamma \cdot (\gamma')^{-1}} \omega_i = \int_{\sigma} \omega_i.$$

Hence

$$\left(\int_{\gamma} \omega_1, \dots, \int_{\gamma} \omega_q \right) - \left(\int_{\gamma'} \omega_1, \dots, \int_{\gamma'} \omega_q \right) = \varphi([\sigma]) \in \Lambda.$$

Since $\text{alb}_X(x)$ is by definition the class of $(\int_{x_0}^x \omega_1, \dots, \int_{x_0}^x \omega_q)$ in $\mathbb{C}^q/\Lambda$, the two choices of path yield vectors in $\mathbb{C}^q$ differing by an element of Λ , hence define the same class in $\mathbb{C}^q/\Lambda$. Thus $\text{alb}_X(x)$ does not depend on the path chosen, and alb_X is a well-defined map $X \rightarrow \text{Alb}(X)$. ■

2.2 Universal property of Albanese mapping

The Albanese mapping enjoys a universal property.

Theorem 2.4 (Universal property of the Albanese map, [Huy05]). *Let $\varphi : X \rightarrow T$ be a holomorphic map from a compact complex manifold X (with base point x_0) to a complex torus T . There is a unique homomorphism of complex tori $\tilde{\varphi} : \text{Alb}(X) \rightarrow T$ such that*

$$\varphi(x) = \tilde{\varphi}(\text{alb}_X(x)) + \varphi(x_0), \quad x \in X.$$

$$\begin{array}{ccc} X & \xrightarrow{\text{alb}_X} & \text{Alb}(X) \\ & \searrow \varphi & \downarrow \tilde{\varphi} + \varphi(x_0) \\ & & T \end{array}$$

Proof. Write $T = \mathbb{C}^g/\Lambda$ for a lattice $\Lambda \subset \mathbb{C}^g$. Let $\pi : \tilde{X} \rightarrow X$ be the universal cover, with $\pi_1(X, x_0)$ acting as the deck transformation group, and fix a lift $\tilde{x}_0 \in \tilde{X}$ of x_0 . Recall

$$\text{Alb}(X) = H^0(X, \Omega_X^1)^\vee / H_1(X, \mathbb{Z}), \quad \text{alb}_X(x) = \left(\omega \mapsto \int_{x_0}^x \omega \right) \bmod H_1(X, \mathbb{Z}).$$

Since $\tilde{X}$ is simply connected, the map $\varphi \circ \pi : \tilde{X} \rightarrow T$ lifts along the universal cover $\mathbb{C}^g \rightarrow T$ to a holomorphic map

$$\phi = (\phi_1, \dots, \phi_g) : \tilde{X} \rightarrow \mathbb{C}^g, \quad \phi(\tilde{x}_0) = 0.$$

The lifting is equivariant. For $\gamma \in \pi_1(X, x_0)$, both $\phi \circ \gamma$ and ϕ lift $\varphi \circ \pi$ (as $\pi \circ \gamma = \pi$), so they differ by a deck transformation of $\mathbb{C}^g \rightarrow T$:

$$\phi(\gamma \cdot z) - \phi(z) \in \Lambda \quad \text{for all } z \in \tilde{X}. \quad (*)$$

Fix γ and set $f_\gamma := \phi \circ \gamma - \phi$, a holomorphic map $\tilde{X} \rightarrow \mathbb{C}^g$ valued in the discrete set Λ by (*). Continuity of f_γ then forces f_γ to be locally constant. As $\tilde{X}$ is connected, f_γ is a single constant $c_\gamma \in \Lambda$. Differentiating,

$$\gamma^*(d\phi_i) = d\phi_i, \quad i = 1, \dots, g, \quad \gamma \in \pi_1(X, x_0),$$

That is, $d\phi_i$ is a π_1 -invariant holomorphic one-form, so each $d\phi_i$ is the pullback $\pi^*\eta_i$ of a unique $\eta_i \in H^0(X, \Omega_X^1)$.

Fix a basis $\omega_1, \dots, \omega_g$ of $H^0(X, \Omega_X^1)$, $q = h^0(X, \Omega_X^1)$, and write $\eta_i = \sum_{j=1}^q a_{ij} \omega_j$. The matrix $A = (a_{ij})$ defines a linear map

$$A : H^0(X, \Omega_X^1)^\vee \rightarrow \mathbb{C}^g, \quad A(\xi)_i = \xi(\eta_i).$$

Every class in $H_1(X, \mathbb{Z})$ is represented by a loop γ at x_0 . Identifying $[\gamma]$ with the functional $\omega \mapsto \int_\gamma \omega$, and lifting γ to a path from $\tilde{x}_0$ to $\gamma \cdot \tilde{x}_0$ in $\tilde{X}$,

$$A([\gamma])_i = \int_\gamma \eta_i = \int_{\tilde{x}_0}^{\gamma \cdot \tilde{x}_0} d\phi_i = \phi_i(\gamma \cdot \tilde{x}_0) - \phi_i(\tilde{x}_0) \stackrel{(*)}{\in} \Lambda.$$

Hence A descends to a homomorphism of complex tori

$$\tilde{\varphi} : \text{Alb}(X) = H^0(X, \Omega_X^1)^\vee / H_1(X, \mathbb{Z}) \longrightarrow \mathbb{C}^g / \Lambda = T.$$

For $x \in X$, lift a path from x_0 to x to a path in $\tilde{X}$ from $\tilde{x}_0$ to some $\tilde{x}$ over x . Then

$$\varphi(x) - \varphi(x_0) = \phi(\tilde{x}) - \phi(\tilde{x}_0) \bmod \Lambda = \left(\int_{x_0}^x \eta_1, \dots, \int_{x_0}^x \eta_g \right) \bmod \Lambda = A(\text{alb}_X(x)) = \tilde{\varphi}(\text{alb}_X(x)),$$

using the fundamental theorem of calculus on each component and the definition of A . This gives

$$\varphi(x) = \tilde{\varphi}(\text{alb}_X(x)) + \varphi(x_0), \quad x \in X.$$

■

The Albanese mapping has a relative version, which will be used in [DPS94].

Theorem 2.5. *Let X be a compact manifold in $\mathcal{C}$, and let $f : X \rightarrow Y$ be a fiber space. Then an Albanese reduction of f exists and is unique up to bimeromorphic equivalence. Here $A(X/Y) \rightarrow Y$ is a torus fibration whose fibers are $\text{Alb}(X_y)$.*

$$\begin{array}{ccc} X & \xrightarrow{\quad} & A(X/Y) \\ & \searrow & \swarrow \\ & Y & \end{array}$$

The restriction to a every fiber X_y gives the Albanese reduction of X_y .

2.3 The Albanese generation lemma

In general the Albanese map need not have connected fibers, nor need it be surjective. Although it is not surjective in general, its image nonetheless generates the Albanese variety; we call this the Albanese Generation Lemma. We then give several applications.

Proposition 2.6 ([Uen75, Lemma 9.14], Albanese Generation Lemma). *Let V be a compact connected complex manifold. The image $\text{alb}(V)$ generates the torus $\text{Alb } V$ as a group. More precisely, for sufficiently large k , the morphism*

$$\text{alb}_V^k : V^k \rightarrow \text{Alb } V, \quad (z_1, \dots, z_k) \mapsto \sum_i \text{alb}_V(z_i),$$

where the right-hand sum is taken using the group structure of the torus $\text{Alb } V$, is surjective.

Proof. Due to limited time, we omit the complete proof and only sketch the idea. After translating the base point, we may assume that there exists a point $z \in V$ with $\text{alb}_V(z) = 0$. For each $k \geq 1$ the image gives a subvariety $A_k \subset \text{Alb } V$, and these form a nested sequence

$$A_1 \subset A_2 \subset \dots \subset A_n \subset \dots$$

Since each A_k is irreducible^a, there exists an n_0 such that

$$A_{n_0} = A_{n_0+1} = \dots = A.$$

The difficult part is to show that A is a subtorus (i.e. stable under the group law), taking this for granted.

We can end the proof by showing that $\text{Alb}(V) = A$ using the universal property of the Albanese mapping. Indeed, let $g : V \rightarrow B$ be a morphism into a complex torus B ; then by the universal property (Theorem 2.4), g factors through the Albanese mapping $\text{alb}_V : V \rightarrow \text{Alb}(V)$. Since the image of the Albanese mapping is contained in A , we have the following commutative diagram.

$$\begin{array}{ccc} V & \xrightarrow{g} & B \\ & \searrow & \nearrow h \\ & \text{Alb } V & \\ \text{alb}_V \swarrow & \uparrow & \nearrow \\ & A & \end{array}$$

Since A is a complex torus, the universal property (Theorem 2.4) again shows that A is the Albanese torus of V , that is, $A = \text{Alb}(V)$. ■

^{“YL: here use the general fact that if $X \subset Y$ are irreducible subvarieties, then either $X = Y$ or $\dim X < \dim Y$; see e.g. Vakil’s FOAG 12.1.5, whose proof also works in the analytic case.}

As a direct consequence of the previous proposition, we see that for a Moishezon manifold the Albanese torus is an abelian variety. For a general compact Kähler manifold, however, the Albanese variety need not be projective.

Corollary 2.7 ([Uen75, Proposition 9.15]). *If X is a Moishezon manifold (in particular, if X is projective), then $\text{Alb}(X)$ is an abelian variety.*

Proof. There is an integer $n > 0$ such that the n -fold sum map

$$\text{alb}_X^n : X \times \cdots \times X \longrightarrow \text{Alb}(X), \quad (z_1, \dots, z_n) \mapsto \text{alb}_X(z_1) + \cdots + \text{alb}_X(z_n),$$

is surjective: the translates of $\text{alb}_X(X)$ generate $\text{Alb}(X)$ as a group. Since X is Moishezon, so is the product $X \times \cdots \times X$. The map alb_X^n is a proper surjective holomorphic map from this Moishezon manifold onto $\text{Alb}(X)$, and a proper surjective holomorphic image of a Moishezon space is again Moishezon; hence $\text{Alb}(X)$ is Moishezon.

Finally, a compact complex torus is Moishezon if and only if it is an abelian variety, since a compact complex torus is always Kähler and a Kähler Moishezon manifold is always projective. ■

Using the Albanese generation lemma, we can also give a criterion for the Albanese map to be surjective in terms of the Kodaira dimension of its image. We first state, without proof, a lemma on the Kodaira dimension of subvarieties of a complex torus.

Lemma 2.8 (Kodaira dimension of subvarieties of complex torus, [Uen75, Corollary 10.5]).

Let $B \subset A$ be a subvariety of a complex torus. Then:

- (1) $\kappa(B) \geq 0$,
- (2) $\kappa(B) = 0$ if and only if B is a translate of a subtorus of A ;
- (3) if B generates A as a group (under translations), then $\kappa(B) > 0$.

Corollary 2.9 ([Uen75, Corollary 10.6]). *Let $\text{alb}_V : V \rightarrow \text{Alb}(V)$ be the Albanese mapping. Then $\kappa(\text{alb}_V(V)) \geq 0$, with equality to 0 if and only if alb_V is surjective.*

Proof. Let $Z = \text{alb}_V(V) \subseteq \text{Alb}(V)$. By the Albanese generation lemma, Z generates $\text{Alb}(V)$ as a group. If alb_V is not surjective, Z is a proper subvariety generating $\text{Alb}(V)$, so $\kappa(Z) > 0$ by Lemma 2.8. If alb_V is surjective, $Z = \text{Alb}(V)$ has trivial canonical bundle, so $\kappa(Z) = 0$. In either case $\kappa(Z) \geq 0$, with equality precisely when alb_V is surjective. ■

A variety with maximal Albanese dimension and canonical singularities has non-negative Kodaira dimension.

Lemma 2.10. *Let X be a Kähler variety with maximal Albanese dimension, then the Kodaira dimension $\kappa(X) \geq 0$.*

Proof. Since the Kodaira dimension is a birational invariant for varieties with canonical singularities, we may assume that X is smooth. Since X has maximal Albanese dimension, hence $\kappa(X) = \kappa(\text{alb}_X(X)) \geq 0$, where ≥ 0 is because a subvariety of the complex torus $A(X)$ has non-negative Kodaira dimension we mentioned earlier. ■

2.4 Albanese mapping for varieties with rational singularities

For applications to the minimal model program one needs the Albanese map for varieties with mildly singular (not necessarily smooth) varieties. We first show a birational invariance property of the Albanese mapping.

Lemma 2.11 ([Uen75, Lemma 9.11]). *A meromorphic map from a compact complex manifold to a complex torus is automatically a morphism (holomorphic everywhere, with no points of indeterminacy).*

Proof. Take a common resolution of the meromorphic map $X \dashrightarrow Y$, given by $p : W \rightarrow X$ and $q : W \rightarrow Y$. Suppose the exceptional set of p is non-empty. By Abhyankar's lemma there exists a rational curve $C \subset \text{Ex}(p)$, which is therefore contracted by p . On the other hand Y contains no rational curve, so C is contracted by q as well which is impossible for a curve in $X \times Y$. Hence the exceptional set of p is empty, and $X \dashrightarrow Y$ is in fact a morphism. ■

Proposition 2.12 (Birational invariance; [Uen75, Proposition 9.12]). *If $f : \widehat{X} \rightarrow X$ is a bimeromorphic morphism of complex manifolds, then $(\text{Alb}(X), \text{alb}_X \circ f)$ is the Albanese torus of $\widehat{X}$. Consequently, having maximal Albanese dimension is a birational invariant, and one may speak of the Albanese dimension of a singular or non-complete variety via any (equivalently, every) resolution.*

Proof. Let $g : X \dashrightarrow \widehat{X}$ be the inverse of f . Fix $\widehat{x}_0 \in \widehat{X}$ and translate alb_X and $\text{alb}_{\widehat{X}}$ by constants so that $\text{alb}_{\widehat{X}}(\widehat{x}_0)$ and $\text{alb}_X(f(\widehat{x}_0))$ are the respective origins.

By Lemma 2.11, the meromorphic map $\text{alb}_X \circ f : \widehat{X} \rightarrow \text{Alb}(X)$ is in fact a morphism, so the universal property (Theorem 2.4) produces a homomorphism of complex tori

$$A(f) : \text{Alb}(\widehat{X}) \rightarrow \text{Alb}(X), \quad \text{alb}_X \circ f = A(f) \circ \text{alb}_{\widehat{X}}.$$

Symmetrically, $\text{alb}_{\widehat{X}} \circ g : X \dashrightarrow \text{Alb}(\widehat{X})$ is, again by Lemma 2.11, a genuine morphism, giving a homomorphism

$$A(g) : \text{Alb}(X) \rightarrow \text{Alb}(\widehat{X}), \quad \text{alb}_{\widehat{X}} \circ g = A(g) \circ \text{alb}_X.$$

Since $f \circ g = \text{id}_X$ and $g \circ f = \text{id}_{\widehat{X}}$ as bimeromorphic maps, substituting the two displayed identities into each other gives $A(f) \circ A(g) = \text{id}$ and $A(g) \circ A(f) = \text{id}$ on the dense open subsets where the relevant compositions are defined, and hence everywhere. So $A(f)$ is an isomorphism, and $(\text{Alb}(X), \text{alb}_X \circ f)$ satisfies the universal property of the Albanese torus for $\widehat{X}$. ■

Definition 2.13 (Albanese map for singular varieties). Let X be a compact normal variety in Fujiki class $\mathcal{C}$. Let $\mu : Y \rightarrow X$ be a Kähler resolution of X . We then define the Albanese mapping as

$$\text{alb}_Y \circ \mu^{-1} : X \dashrightarrow \text{Alb}(Y).$$

This is independent of the choice of resolution by [Proposition 2.12](#), hence well defined.

Typically, we will work with Fujiki varieties with rational singularities, as in this case the Albanese mapping is holomorphic.

Proposition 2.14 ([BS95, Lemma 2.4.1]). Let X be a complete normal variety in Fujiki's class $\mathcal{C}$ with at worst rational singularities. Then the Albanese map $X \dashrightarrow \text{Alb}(X)$ defined via a resolution is in fact an everywhere-defined morphism $\text{alb}_X : X \rightarrow \text{Alb}(X)$, and coincides with the natural map $X \rightarrow \text{Pic}^0(X)^\vee$ induced by the Poincaré line bundle.

Before proving this proposition, let us say a few words about the natural holomorphic map $X \rightarrow \text{Pic}^0(X)^\vee$. Recall that, given a compact normal variety in Fujiki class $\mathcal{C}$, there is a universal line bundle $\mathcal{P}$ on $X \times \text{Pic}^0(X)$ such that

$$\mathcal{P}|_{X \times [L]} \cong L, \quad \mathcal{P}|_{\{x_0\} \times \text{Pic}^0(X)} \cong \mathcal{O}_{\text{Pic}^0(X)}.$$

We can then consider the restriction $\mathcal{P}|_{\{x\} \times \text{Pic}^0(X)} \in \text{Pic}(\text{Pic}^0(X)) = \text{Pic}^0(X)^\vee$, which clearly varies holomorphically in x and hence defines a holomorphic map

$$X \rightarrow \text{Pic}^0(X)^\vee, \quad x \mapsto \mathcal{P}_x.$$

Proof. Let $\mu : Y \rightarrow X$ be a resolution and then we get a meromorphic map

$$X \dashrightarrow \text{Alb}(Y),$$

which is by definition the albanese mapping associated to X . By Leray spectral sequence argument, rational singularities give $H^1(X, \mathcal{O}_X) \cong H^1(Y, \mathcal{O}_Y)$ via μ^* , from which one deduces that $\mu^* : \text{Pic}^0(X) \rightarrow \text{Pic}^0(Y)$ is étale^a and injective, hence μ^* is an isomorphism (étale morphism in degree 1 is isomorphism). Since Y is smooth and Kähler, we have

$$\text{Alb}(Y) \cong \text{Pic}^0(Y)^\vee \cong \text{Pic}^0(X)^\vee.$$

Hence the Albanese map is nothing but the natural holomorphic map

$$X \rightarrow \text{Pic}^0(X)^\vee$$

mentioned above. ■

^aWhere étaleness is because, take the differential to get

$$d\mu^* : H^1(X, \mathcal{O}_X) \rightarrow H^1(Y, \mathcal{O}_Y),$$

which is an isomorphism, here the tangent space (at identity) of the Picard variety can be identified with $H^1(X, \mathcal{O}_X)$. On the other hand, since μ is birational hence is dominant, therefore the pull back μ^* is injective.

3 Irregularity and Albanese dimension

We now introduce the numerical invariants attached to $\text{Alb}(X)$ and alb_X , following Ueno's book [Uen75], which is the standard reference for this material.

Definition 3.1 (Irregularity; [Uen75, Definition 9.20]). Let V be a complex variety with non-singular model V^* . The irregularity of V is

$$q(V) := \dim_{\mathbb{C}} H^1(V^*, \mathcal{O}_{V^*}),$$

independent of the choice of resolution V^* and a bimeromorphic invariant of V .

Definition 3.2 (Albanese dimension; [Uen75, Definition 9.21]). The Albanese dimension of a complex variety V is $a(V) := \dim \text{Alb}(V^*)$, the dimension of the Albanese torus of a non-singular model V^* . We say V has maximal Albanese dimension if $a(V) = \dim V$, equivalently the Albanese map is generically finite onto its image.

Definition 3.3 (Augmented irregularity).

$$\widehat{q}(X) := \sup\{q(X') \mid X' \rightarrow X \text{ a finite étale cover}\} \in \mathbb{Z}_{\geq 0} \cup \{\infty\}.$$

Proposition 3.4 (Behaviour under surjective morphisms, [Uen75, Lemma 9.23]). Let $f : V \rightarrow W$ be a surjective morphism of complex varieties. Then

$$q(V) \geq q(W), \quad a(V) \geq a(W).$$

Proof. Pullback of forms $f^* : H^0(W, \Omega_W^1) \rightarrow H^0(V, \Omega_V^1)$ is injective because f is dominant, which gives $q(V) \geq q(W)$. The Albanese-dimension inequality follows by applying this together with the universal property (Theorem 2.4): the composite $V \rightarrow W \rightarrow \text{Alb}(W)$ factors through $\text{Alb}(V)$, so $\text{Alb}(W)$ is (a quotient of) a further quotient torus of $\text{Alb}(V)$. ■

We next prove a subadditivity inequality using the E_2 -degeneration of Leray spectral sequence. This relation will be important in the next section.

Lemma 3.5 (Subadditivity of irregularity). *Let $g : X \rightarrow Y$ be a smooth fibration with connected fibres between compact Kähler manifolds, with fibre F . Then*

$$q(X) \leq q(Y) + q(F),$$

with equality if and only if the monodromy representation $\pi_1(Y) \rightarrow \text{Aut}(H^1(F, \mathbb{Z}))$ is trivial.

Similar to the Iitaka conjecture, the subadditive direction requires no extra hypothesis, whereas for the additive relation a Hodge-theoretic condition appears. Later we will upgrade this condition to augmented irregularity.

Proof. Consider the Leray spectral sequence of the constant sheaf $\mathbb{R}_X$ for g ,

$$E_2^{s,t} = H^s(Y, R^t g_* \mathbb{R}_X) \implies H^{s+t}(X, \mathbb{R}).$$

Since the fibres are connected and g is (differentiably) a locally trivial fibre bundle, $R^0 g_* \mathbb{R}_X = \mathbb{R}_Y$, and $R^1 g_* \mathbb{R}_X$ is a real local system of rank $2q(F)$ containing $R^1 g_* \mathbb{Z}_X$ as a lattice, so its monodromy lies in $SL_{2q(F)}(\mathbb{Z})$. Indeed $H^1(F, \mathbb{Z})$ is a free lattice of rank $b_1 = 2q(F)$, so the monodromy lies in $GL(H^1(F, \mathbb{Z})) = GL_{2q(F)}(\mathbb{Z})$. On the other hand, for a geometric family the monodromy preserves the intersection pairing, hence actually lies in the smaller subgroup $SL_{2q(F)}(\mathbb{Z})$.

Since the global sections of a local system are isomorphic to the monodromy-invariant part of the fiber, we get

$$\dim H^0(Y, R^1 g_* \mathbb{R}_X) \leq 2q(F),$$

with equality iff the monodromy is trivial. By Blanchard's theorem, the Leray spectral sequence of a Kähler fibration degenerates at E_2^a , so

$$\begin{aligned} 2q(X) &= \dim H^1(X, \mathbb{R}) \\ &= \dim H^1(Y, \mathbb{R}) + \dim H^0(Y, R^1 g_* \mathbb{R}_X) \\ &\leq 2q(Y) + 2q(F), \end{aligned}$$

with equality iff the monodromy representation $\pi_1(Y) \rightarrow \text{Aut}(H^1(F, \mathbb{Z}))$ is trivial. ■

^aOne should be careful: the Kähler condition is vital for the spectral-sequence argument; a non-Kähler smooth fibration may fail to have E_2 -degeneration, e.g. the Hopf fibration.

Lemma 3.6 (Subadditive for augmented irregularity). *In the situation of Lemma 3.5, $\widehat{q}(X) \leq \widehat{q}(Y) + \widehat{q}(F)$.*

Proof. The homotopy exact sequence of the fibration gives

$$\cdots \rightarrow \pi_2(Y) \rightarrow \pi_1(F) \rightarrow \pi_1(X) \rightarrow \pi_1(Y) \rightarrow 1.$$

A finite étale cover $\widetilde{X}$ of X corresponds to a finite-index subgroup $\pi_1(\widetilde{X}) \subset \pi_1(X)$; its

image in $\pi_1(Y)$ and preimage in $\pi_1(F)$ are again finite-index, giving finite étale covers $\tilde{Y}, \tilde{F}$ fitting into a smooth fibration $\tilde{X} \rightarrow \tilde{Y}$ with fibre $\tilde{F}$.

$$\begin{array}{ccc} \tilde{F} & \longrightarrow & F \\ \downarrow & & \downarrow \\ \tilde{X} & \longrightarrow & X \\ \downarrow & & \downarrow \\ \tilde{Y} & \longrightarrow & Y \end{array}$$

By Lemma 3.5,

$$q(\tilde{X}) \leq q(\tilde{Y}) + q(\tilde{F}) \leq \widehat{q}(Y) + \widehat{q}(F);$$

taking the supremum over $\tilde{X}$ gives the claim. ■

We next prove a rigidity theorem for torus fibrations using Hodge theory.

Lemma 3.7 (Rigidity of Kähler torus fibrations). *A smooth Kähler fibration by complex tori*

$$f : X \rightarrow \mathbb{C}$$

is holomorphically trivial.

Proof. By a result of Griffiths and Schmid, the classifying (period) map $\mathbb{C} \rightarrow D$ into the Siegel upper half space the moduli space of complex tori polarized by a Kähler form ω , with given periods is always constant, by the Brody hyperbolicity of D . ■

Proposition 3.8 (Demailly–Peternell–Schneider’s addition formula for irregularity, [DPS94, Proposition 3.12]). *Let $g : X \rightarrow Y$ be a smooth fibration with connected fibres between compact Kähler manifolds, with fibre F . Then*

$$q(X) \leq q(Y) + q(F), \quad \widehat{q}(X) \leq \widehat{q}(Y) + \widehat{q}(F).$$

If moreover Y is a complex torus and $\pi_1(F)$ is virtually abelian with the Albanese map of every finite étale cover of F of constant rank, then $\widehat{q}(X) = \widehat{q}(Y) + \widehat{q}(F)$.

Proof. We only highlight the idea of the proof. Consider the relative Albanese factorization $X \rightarrow A(X/Y) \rightarrow Y$, a smooth torus fibration with fibres $A(F_y)$. We show that $A(X/Y) \rightarrow Y$ is a holomorphically locally trivial torus bundle, using the rigidity result proved above: since Y is a complex torus, its universal cover is $\mathbb{C}^n$, which is why we can apply the Griffiths–Schmid rigidity result^a. Therefore, for a small $U \ni y$, on $g^{-1}(U)$ there are holomorphic maps over U

$$g^{-1}(U) \rightarrow A(g^{-1}(U)/U) \simeq A(F_y) \times U \xrightarrow{\text{pr}_2} A(F_y).$$

Recall that we already showed that

$$\rho : \pi_1(Y) \rightarrow GL(H^1(F_y, \mathbb{R})) = GL_{2q}(\mathbb{R})$$

has image $\rho(\pi_1(Y)) \subseteq SL_{2q}(\mathbb{Z})$, a discrete subgroup. We now introduce a compactness constraint, i.e., the monodromy group lies in some orthogonal group: we construct a metric such that

$$\langle \rho(\gamma)u, \rho(\gamma)v \rangle_y = \langle u, v \rangle_y.$$

To do this, consider the holomorphic 1-form on F_y and define the usual

$$\langle u, v \rangle_y = \int_{F_y} iu \wedge \bar{v} \wedge \omega^{p-1} > 0.$$

Since the torus fibration is holomorphically trivial, we may take holomorphic extension take u', v' to be d -closed holomorphic 1-forms on all of $g^{-1}(U)$; restricting to each fibre $F_{y'}$, define

$$\langle u', v' \rangle_{y'} = \int_{F_{y'}} \Theta, \quad \Theta := iu' \wedge \bar{v}' \wedge \omega^{p-1}, \quad p = \dim F_{y'}.$$

By Stokes' formula we can check

$$\int_{F_{y_1}} \Theta - \int_{F_{y_0}} \Theta = \int_{g^{-1}(\gamma)} d\Theta = 0,$$

hence it is independent of y .

In summary, for the Kähler fibration $X \rightarrow Y$ with fibre F , the monodromy

$$\pi_1(Y) \rightarrow \text{Aut}(H^1(F', \mathbb{Z}))$$

has finite image, since it is contained in $SL_{2q}(\mathbb{Z}) \cap O_{2q}(\langle, \rangle)$. Hence, after a finite cover giving a fibration $\tilde{X} \rightarrow \tilde{Y}$ with fibre $\tilde{F}$ and trivial monodromy, we apply [Lemma 3.5](#) to deduce the equality. ■

^aTo apply the Griffiths–Schmid result we must check that $A(X/Y)$ is Kähler. Please check [\[DPS94\]](#)

4 Albanese fibrations for Kähler manifolds with nef tangent bundle

We now turn to Demailly–Peternell–Schneider's classification of compact Kähler manifolds with numerically effective (nef) tangent bundle, in which the Albanese fibration is the central structural device. Their remarkable work shows that for a compact Kähler manifold with T_X nef, alb_X is a smooth fibration whose fibres are Fano (hence simply connected). First we show a result particularly suited to the induction.

Theorem 4.1 (Demailly–Peters–Schneider first structure theorem, Proposition 3.9). *Let X be a Kähler manifold with nef tangent bundle T_X , and let $\text{alb}_X: X \rightarrow A(X)$ be the Albanese map, where $A(X) = \mathbb{C}^q/\Lambda$ is the Albanese torus. Then:*

1. alb_X is a submersion;
2. the relative tangent bundle $T_{X/A(X)}$ is nef;
3. the fibres of alb_X are connected and have nef tangent bundle.

Proof. Step 1: alb_X is a submersion. The torus $A(X) = \mathbb{C}^q/\Lambda$ carries translation-invariant holomorphic 1-forms $dz_1, \dots, dz_q$. Pulling back via alb_X gives global sections

$$u_i = \text{alb}_X^*(dz_i) = d(\text{alb}_X)_i \in H^0(X, \Omega_X^1), \quad i = 1, \dots, q,$$

so that the differential of alb_X is encoded by

$$d\text{alb}_X = (d(\text{alb}_X)_1, \dots, d(\text{alb}_X)_q): T_X \longrightarrow \text{alb}_X^* T_{A(X)}.$$

Suppose alb_X fails to be a submersion at some point $x \in X$, i.e. $\text{rk}(d\text{alb}_X(x)) < q$. Then the vectors $(d(\text{alb}_X)_i(x))_{i=1, \dots, q}$ are linearly dependent, so there is a nonzero linear combination

$$u = \lambda_1 u_1 + \dots + \lambda_q u_q \in H^0(X, \Omega_X^1)$$

with $u(x) = 0$.

Since T_X is nef, its dual bundle $\Omega_X^1 = T_X^*$ is a dual of a nef vector bundle. Since a nontrivial section of the dual of a nef vector bundle vanishes nowhere, u cannot vanish at any point of X . This contradicts $u(x) = 0$. Hence $\text{rk}(d\text{alb}_X) = q$ everywhere, and alb_X is a submersion.

Step 2: the fibres have nef tangent bundle. Since alb_X is a submersion, we have the short exact sequence of vector bundles on X

$$0 \longrightarrow T_{X/A(X)} \longrightarrow T_X \xrightarrow{d\text{alb}_X} \text{alb}_X^* T_{A(X)} \longrightarrow 0.$$

As $A(X)$ is a torus, $T_{A(X)}$ is trivial, hence $\text{alb}_X^* T_{A(X)}$ is nef (indeed trivial). Since T_X is nef and quotients of nef bundles are nef, while the kernel of a surjection from a nef bundle onto a nef bundle is again nef, it follows that $T_{X/A(X)}$ is nef. Restricting to a fibre $F = \text{alb}_X^{-1}(y)$ gives $T_{X/A(X)}|_F = T_F$, so every fibre has nef tangent bundle.

Step 3: the fibres are connected. Suppose some fibre of alb_X is disconnected. Taking connected components of fibres yields the Stein factorization

$$\text{alb}_X: X \xrightarrow{f} Y \xrightarrow{g} A(X),$$

where f has connected fibres and, since alb_X is a submersion, f is a smooth fibration; hence g is a finite étale covering^a. We may assume $\deg(g) > 1$, since otherwise g is an isomorphism and the fibres of alb_X are already connected.

Because $g: Y \rightarrow A(X)$ is a finite étale cover of a complex torus, Y is itself a complex torus. By the universal property of the Albanese map applied to $f: X \rightarrow Y$, there is a factorization

$$X \xrightarrow{\text{alb}_X} A(X) \xrightarrow{h} Y$$

with $g \circ h = \text{id}_{A(X)}$ and $h \circ g = \text{id}_Y$ up to translation, by the uniqueness part of the universal property of complex tori applied to the composite $A(X) \rightarrow Y \rightarrow A(X)$. In particular g is an isomorphism, since g is a finite morphism between abelian varieties of the same dimension whose tangent map is an isomorphism, hence an isogeny of degree 1. This contradicts $\deg(g) > 1$. Therefore the fibres of alb_X are connected, completing the proof. ■

^aThis is a smooth version of Stein factorization, that is if alb_X is smooth, we can assume that g is étale, see this version e.g. in Shin-ichi's paper.

Using a Fano criterion in terms of irregularity we can show further that the fiber of the Albanese fibration are Fano! We omit the proof, and mention that the key ingredient is the additive relation for irregularity we proved in the previous section.

Theorem 4.2 (Demailly–Peternell–Schneider second structure theorem, [DPS94, Theorem 3.14, 3.15]). *Let X be a compact Kähler manifold with T_X nef, and let $\tilde{X} \rightarrow X$ be a finite étale cover realising the maximal irregularity $q = \widehat{q}(X) = q(\tilde{X})$. Then:*

- (i) $\pi_1(\tilde{X}) \cong \mathbb{Z}^{2q}$;
 - (ii) *the Albanese map $\text{alb}_{\tilde{X}}: \tilde{X} \rightarrow \text{Alb}(\tilde{X})$ is a smooth fibration over a q -dimensional torus, with nef relative tangent bundle;*
 - (iii) *most importantly the fibres F of this fibration are Fano manifolds with nef tangent bundle.*
- Consequently $\pi_1(X)$ is, as an abstract group, an extension of a finite group by $\mathbb{Z}^{2q}$.*

5 Birational Geometry of varieties with maximal Albanese dimension

Next we discuss a class of varieties with very nice birational geometry: those of maximal Albanese dimension. Our presentation follows [Fuj13] and [DH24a].

Varieties of maximal Albanese dimension have very good minimal model theory, because the target of the Albanese map is (a subvariety of) an abelian variety, and abelian varieties contain no rational curves. The following lemma is the precise mechanism.

Lemma 5.1 ([Fuj13, Lemma 3.2, 3.3]). *Let (X, Δ) be a log canonical pair and $f: X \rightarrow S$ a surjective morphism of projective varieties, with S containing no rational curves. If $K_X + \Delta$ is f -nef, then it is nef. Consequently, if (X', Δ') is a minimal model of (X, Δ) over S , it is already*

a minimal model of (X, Δ) .

Proof. If $K_X + \Delta$ were not nef, the cone theorem would produce a rational curve $C \subset X$ with $(K_X + \Delta) \cdot C < 0$. Since $K_X + \Delta$ is f -nef, C cannot be f -vertical, so $f(C)$ is a positive-dimensional subvariety of S . But $f(C)$ is the image of $\mathbb{P}^1$, hence a rational curve, contradicting that S has none. The second statement is the same argument applied to $K_{X'} + \Delta'$, which is f -nef by definition of a relative minimal model. ■

Remark 5.2. Abelian varieties, and more generally subvarieties of abelian varieties, contain no rational curves (a non-constant map $\mathbb{P}^1 \rightarrow A$ would lift to a non-constant holomorphic map $\mathbb{P}^1 \rightarrow \mathbb{C}^g$, impossible since Liouville's theorem tells us it must be constant). So Lemma 5.1 applies with S any subvariety of $\text{Alb}(X)$, in particular with $S = a_X(X)$.

Theorem 5.3 (Fujino; [Fuj13, Theorem 4.3]). *Let (X, Δ) be a projective klt pair of maximal Albanese dimension. Then (X, Δ) has a good minimal model.*

Proof. klt singularities are in particular rational, so Proposition 2.14 shows $a_X : X \rightarrow \text{Alb}(X)$ is an everywhere-defined morphism; let $S := a_X(X)$ and $f := a_X : X \rightarrow S$. Since X has maximal Albanese dimension, f is generically finite, hence $K_X + \Delta$ is f -big (a generically finite morphism makes every $\mathbb{R}$ -divisor relatively big). By the relative existence theorem for minimal models of big divisors [BCHM10, Theorem 1.2], there is a minimal model $(X', \Delta') \rightarrow S$ of (X, Δ) over S .

Since $S \subseteq \text{Alb}(X)$, Lemma 5.1 shows (X', Δ') is already an (absolute) minimal model of (X, Δ) : $K_{X'} + \Delta'$ is nef. Finally, by the abundance theorem for varieties of maximal Albanese dimension, (X', Δ') is a good minimal model. ■

A general subvariety of a variety with maximal Albanese dimension again has maximal Albanese dimension.

Lemma 5.4. *Let X be a projective variety with maximal Albanese dimension. Let $\pi : \bar{X} \rightarrow X$ be a resolution and let $\text{alb}_{\bar{X}} : \bar{X} \rightarrow \text{Alb}(\bar{X})$ be the Albanese mapping. Let Y be a subvariety of X . Assume that $Y \not\subset \pi(\text{Exc}(\pi) \cup \text{Exc}(\beta))$, where $\beta : \bar{X} \rightarrow V$ is the Stein factorization of $\text{alb}_{\bar{X}} : \bar{X} \rightarrow \text{alb}_{\bar{X}}(\bar{X})$. Then Y has maximal Albanese dimension.*

We next prove the abundance theorem for varieties with maximal Albanese dimension.

Theorem 5.5 (Abundance conjecture for varieties with maximal Albanese dimension, [Fuj13]). *Let (X, Δ) be a projective klt pair such that X has maximal Albanese dimension. Assume that $K_X + \Delta$ is nef. Then $K_X + \Delta$ is semi-ample.*

Proof. We prove the result by induction on dimension. By taking a small $\mathbb{Q}$ -factorial modification and a terminalization, we may assume that (X, Δ) is a $\mathbb{Q}$ -factorial terminal pair. By Lemma 2.10, since X has maximal Albanese dimension, $\kappa(K_X + \Delta) \geq 0$. The key point is that it suffices to show $\kappa(K_X + \Delta) > 0$. Indeed, once we know this, we argue as in [Kaw85]: there exist nonsingular varieties Y, Z , a birational morphism $\mu : Y \rightarrow X$, and the Iitaka fibration $f : Y \rightarrow Z$ of X , as in the following commutative diagram:

$$\begin{array}{ccc} & Y & \\ \mu \swarrow & & \searrow f \\ X & & Z \end{array}$$

Since $\kappa(K_X + \Delta) > 0$, we have $\dim Z > 0$, so a general fiber F of f has dimension $\dim F < \dim X$; on the other hand it still has maximal Albanese dimension (by Lemma 5.4). We then run the MMP on F to get

$$F \dashrightarrow F',$$

where F' again has maximal Albanese dimension (cf. Proposition 2.12). Since $\dim F' < \dim X$, by induction F' satisfies log abundance; combined with $\kappa(K_F + \Delta_F) = 0$, this implies

$$K_{F'} + \Delta_{F'} \sim_{\mathbb{Q}} 0.$$

We can then deduce the result directly, as in Kawamata [Kaw85] (see the appendix of the previous note for details).

Hence it remains to prove that $\nu(K_X + \Delta) > 0$ implies $\kappa(K_X + \Delta) > 0$ (we call it weak abundance)^a. If $\kappa(K_X) > 0$ then obviously $\kappa(K_X + \Delta) > 0$, so we may assume $\kappa(K_X) = 0$ and argue by contradiction. We apply [Kaw81]: in this case $X \rightarrow \text{alb}_X(X)$ is generically finite with connected fibres, hence birational. Since X has terminal singularities, K_X is effective with support equal to the alb_X -exceptional part, therefore

$$\kappa(X, K_X + \Delta) \geq \kappa(X, \text{alb}_X^*(\text{alb}_X)_*\Delta) = \kappa(A, (\text{alb}_X)_*\Delta).$$

We claim $(\text{alb}_X)_*\Delta \neq 0$. Indeed, if $(\text{alb}_X)_*\Delta = 0$, then Δ is alb_X -exceptional; since K_X is also alb_X -exceptional, $K_X + \Delta$ is effective and alb_X -exceptional. On the other hand, by hypothesis $K_X + \Delta$ is nef, so the negativity lemma forces $K_X + \Delta = 0$, contradicting $\nu(K_X + \Delta) > 0$.

Since $(\text{alb}_X)_*\Delta \neq 0$, then

$$\kappa(X, K_X + \Delta) \geq \kappa(X, \text{alb}_X^*(\text{alb}_X)_*\Delta) = \kappa(A, (\text{alb}_X)_*\Delta) \geq 1,$$

where the last inequality holds because any nonzero effective divisor on an abelian variety has positive Iitaka dimension. ■

^aNote that there is one trivial case when $\nu(K_X + \Delta) = 0$, the abundance trivially hold, hence we do not mention it.

Remark 5.6 (The Kähler case, due to Das–Hacon [DH24a]). Fujino’s argument contains two ingredients: the relative minimal model program for projective morphism [BCHM10], and the abundance theorem for maximal Albanese dimension varieties. The first tool is only available under the projective setting. Das–Hacon [DH24b] prove the same conclusion for compact Kähler klt pairs (X, B) of maximal Albanese dimension, but since neither tool is available as such in the Kähler category, the proof cannot simply run the argument above analytically; instead it replaces the abundance-theorem input with a direct argument via cohomology support loci and generic vanishing (Green–Lazarsfeld–Simpson GV-sheaves) to show the relevant divisors are a_X -exceptional, combined with an induction on the Iitaka fibration to handle $\kappa > 0$. This is substantially longer and uses heavier machinery than Fujino’s projective proof. Das–Hacon also record the corollary that if a_X happens to already be projective with connected fibres, one may run the $(K_X + B)$ -minimal model program directly over $\text{Alb}(X)$ (as in the proof of Theorem 5.3 above) to reach a good minimal model.

6 Further discussion

We end this note by pointing out some further directions, in each case stated without proof.

Remark 6.1 (Conditions for the Albanese map to be a fibration). As mentioned above, in general the Albanese mapping need not be a fibration. There are, however, cases in which it is: (1) if the algebraic dimension $a(X) = 0$, see [Uen75, Lemma 13.6]; (2) If the image of alb_X is a curve C , then C is smooth of genus $\dim \text{Alb}(X)$ with connected fibres [Uen75, Proposition 9.19]; (3) When $\kappa(X) = 0$, then the Albanese map has connected fiber, see [Kaw81].

Remark 6.2 (Albanese fibrations in abundance conjectures). Using Kawamata’s structure theorem for finite covers of abelian varieties [Kaw81] to reduce the Albanese map to a fibration over an abelian variety, Nakayama proved that $\kappa_\sigma(K_X + \Delta) = 0$ implies $\kappa(K_X + \Delta) = 0$ for klt pairs [Nak04, Corollary IV.4.9]. Nakayama’s idea goes as follows. Let us first denote $D := K_X + \Delta$. Since $\kappa_\sigma(D) = 0$, so that $P_\sigma(D) \equiv 0$ and hence $D \equiv N_\sigma(D)$. If $q(X) = 0$, then $\text{Pic}^0(X) = 0$, hence $D \sim_{\mathbb{Q}} N_\sigma(D) \geq 0$ we are done. Hence, we may assume $q(X) > 0$. One can prove the Albanese mapping is a fibration $X \rightarrow \text{Alb}(X)$, using $0 = \kappa_\sigma(D)$. And then apply a variant version of easy addition formula

$$0 = \kappa_\sigma(D) \geq \kappa(D; X/Y) + \kappa_\sigma(D, \det \alpha) = \kappa_\sigma(D, \det \alpha) \geq 0,$$

which implies $\kappa_\sigma(D, \det \alpha)$ and then implies non vanishing $\kappa(D) \geq 0$ hence finish the proof. Nakayama contribute the idea of applying addition formula for Albanese fibration to Tsunoda in his book.

Remark 6.3 (Good minimal models for Albanese-fibred varieties). Lai [Lai11] and [Hua26] showed if $\text{alb}_X : X \rightarrow \text{Alb}(X)$ is projective with connected fibres, the general fibre has a good minimal model, then so does X .

Remark 6.4 (Structure of varieties with nef anti-canonical divisor). For (X, Δ) projective klt pair, with $-(K_X + \Delta)$ nef, one can prove that alb_X is a locally constant fibration, see [Cao19, Wan22, CCM21, NW25, iMWWZ25, iMW23]; the Kähler case is still open, the main difficulty being the failure of projectivity of the Albanese fibration.

Remark 6.5 (Ueno's Conjecture K). Ueno conjectured that when $\kappa(X) = 0$, the Albanese map is not merely a fibre space (as above) but becomes, after a finite étale base change $B \rightarrow \text{Alb}(X)$, birationally trivial over B : $X \times_{\text{Alb}(X)} B$ is birational to $F \times B$ over B , where F is the general fibre, with $\kappa(F) = 0$. Ueno's conjecture is a test case for the Iitaka conjecture.

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